Course:
Networked Control

Lecture III.2: Networked control systems – Control networks

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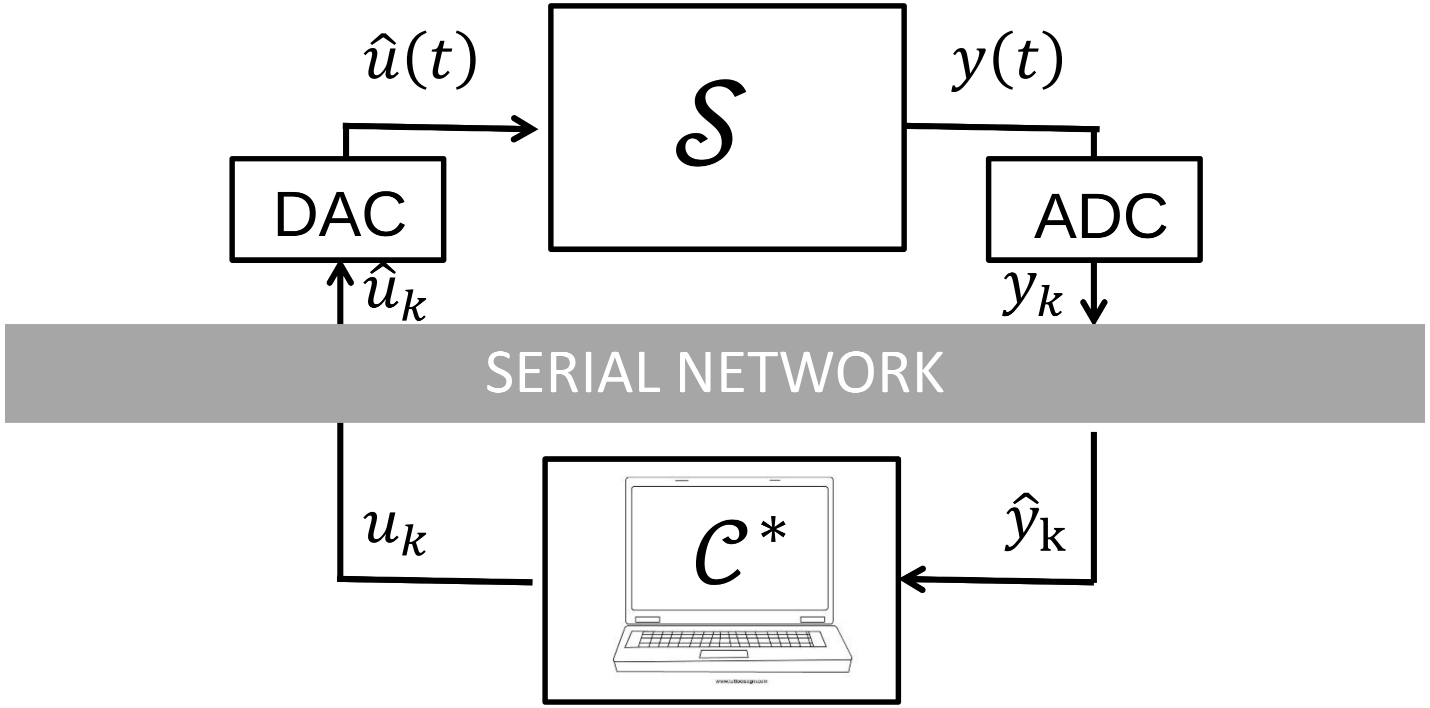
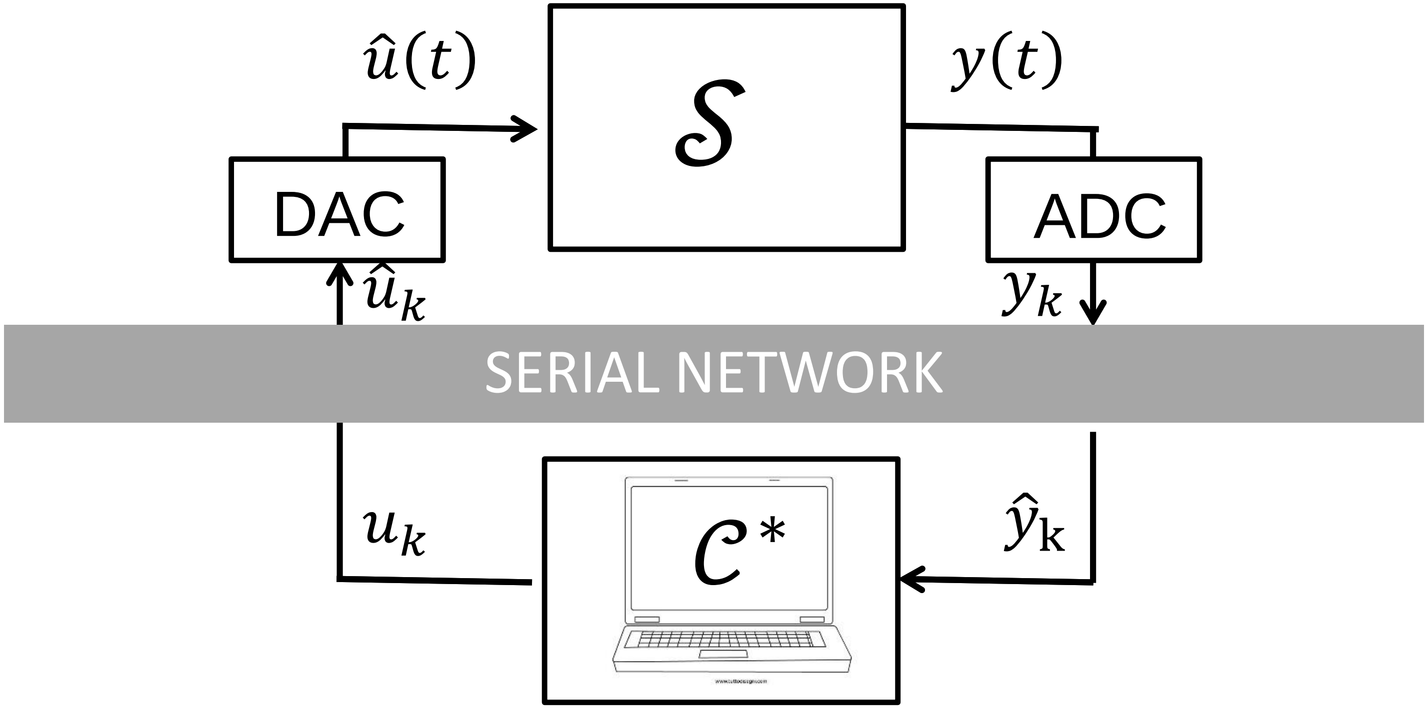
Data networks and control networks

- **Data networks:**
  - Aim: to exactly convey the data, possibly with some delay
  - Large data packets
  - Infrequent bursty information (high-bandwidth transmissions over short periods)
  - Large data rates
- **Control networks**:
  - Aim: support time (and safety) critical and real-time applications
  - Small but frequent exchange of packets between a large set of nodes



Data networks and control networks

The control network impacts on the performance of the control system by creating a difference between the data reconds and the correponding data images.



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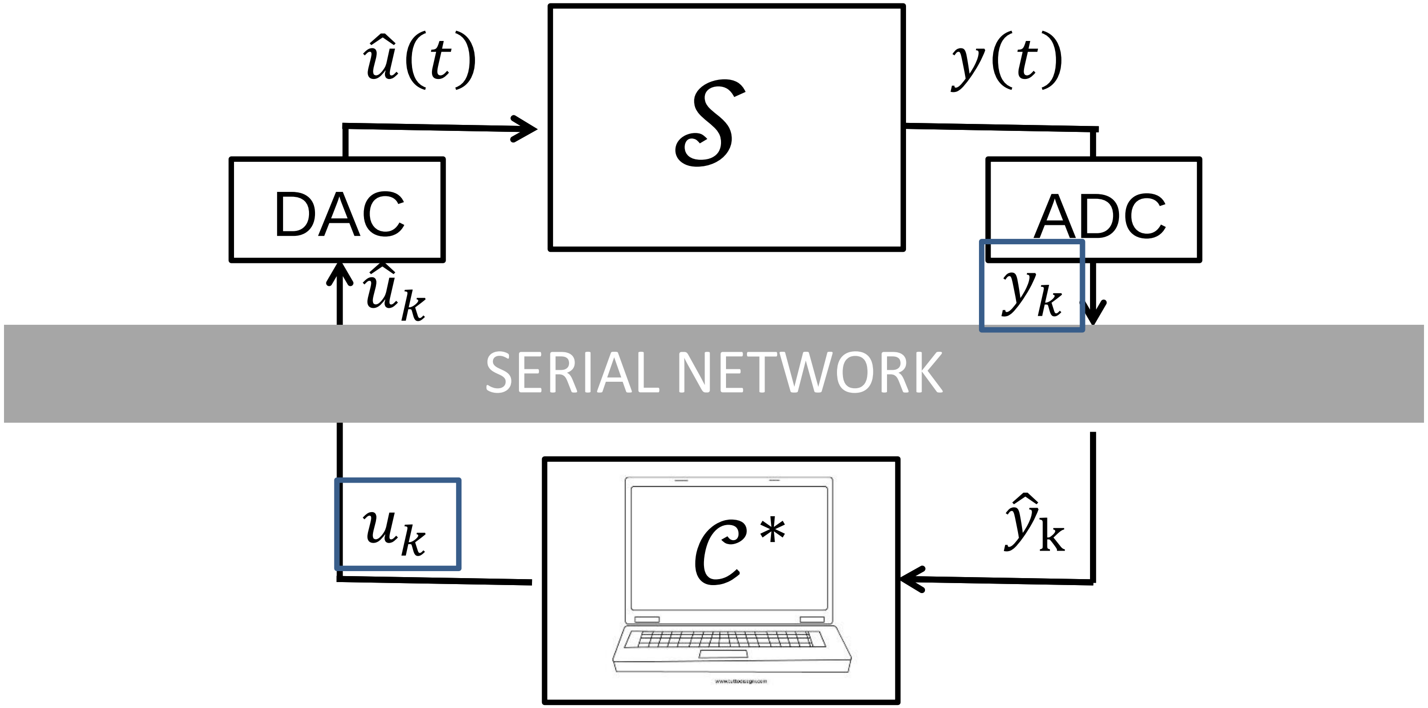

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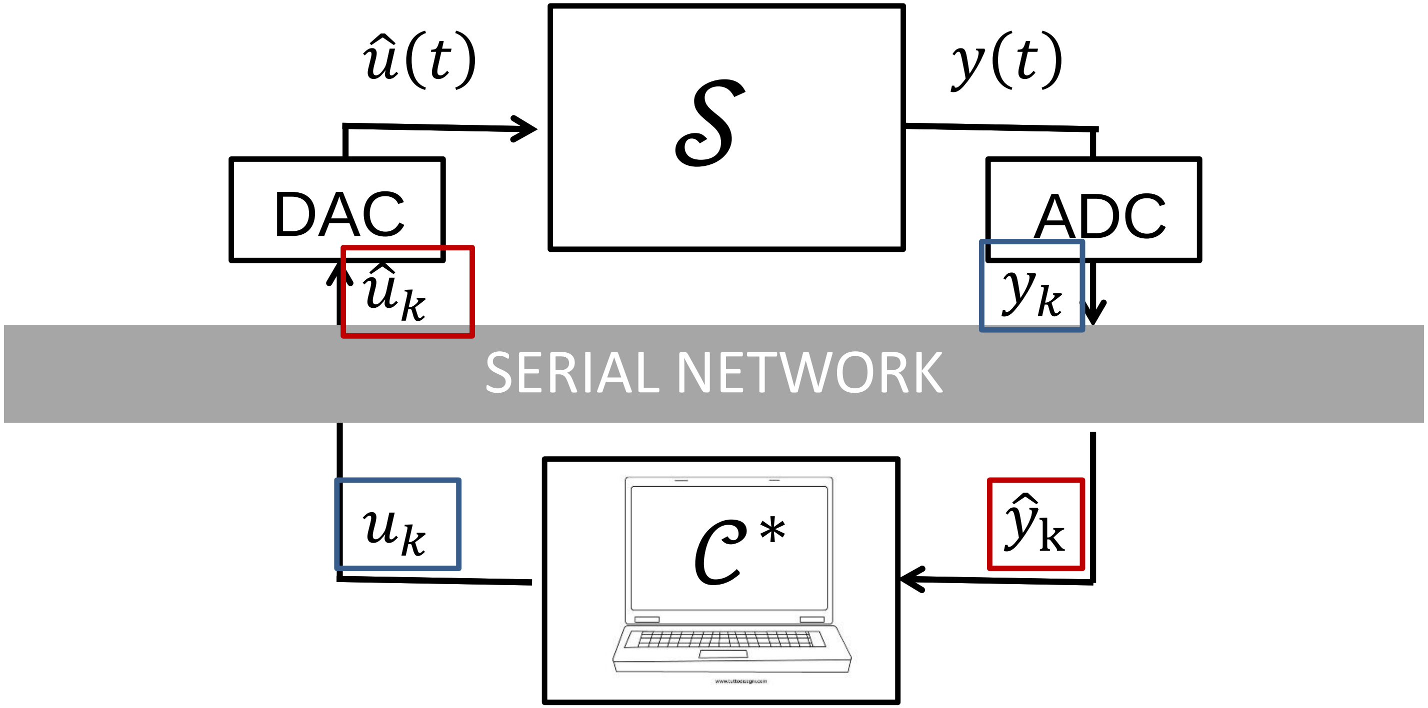

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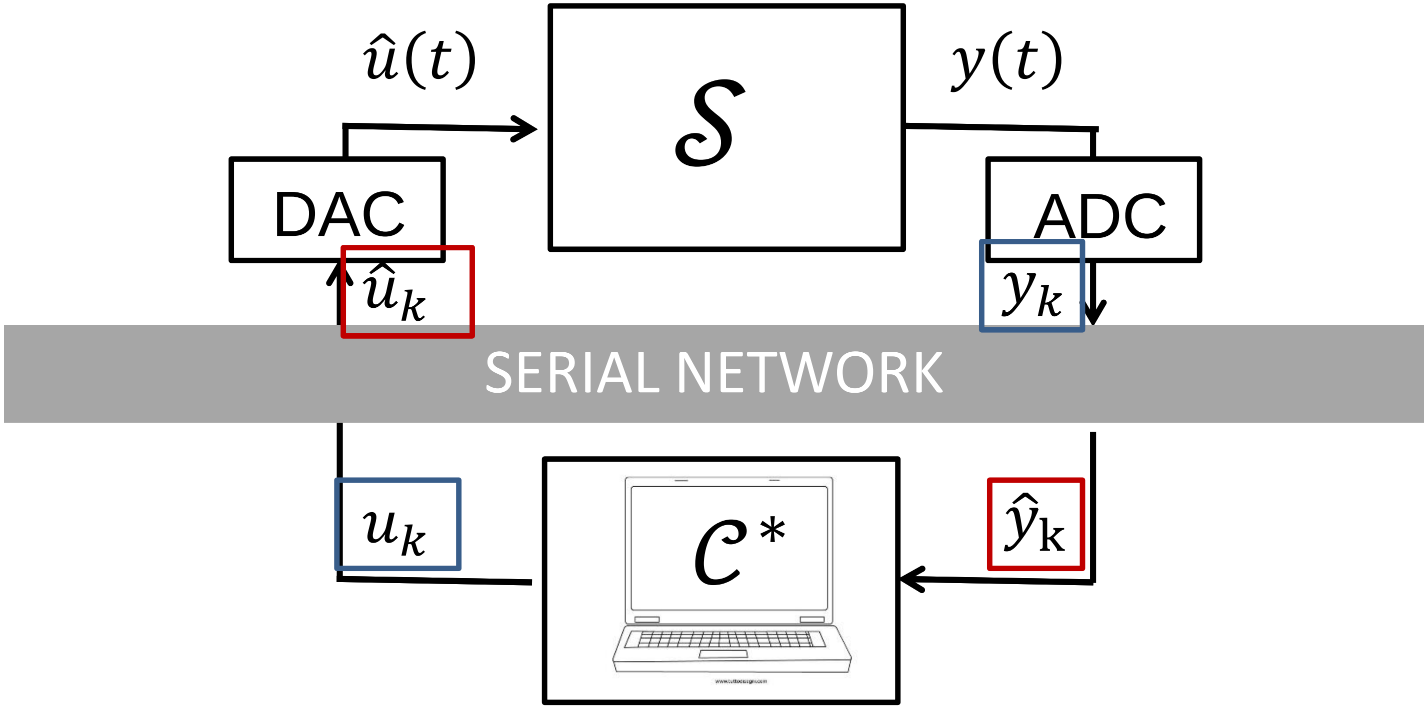
- If the network speed is large and the traffic is sparse, the effect of introducing a network in the feedback loop is that of creating a small, randomly-varying time delay between data records and their images.



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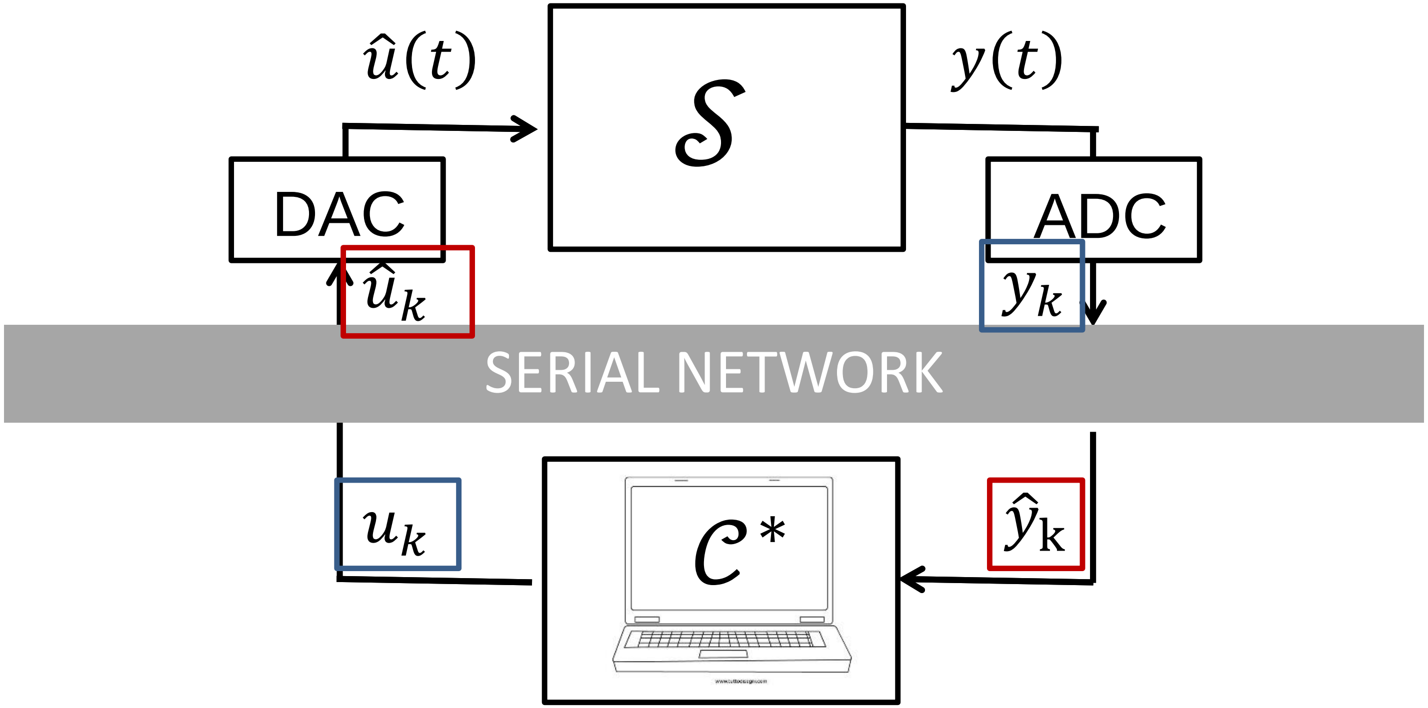
- In presence of congestions, the time spent by a packet waiting to be sent is significant (it may be even greater than the sampling time!)

-> scheduling – MAC media access control – must conveniently manage these situations



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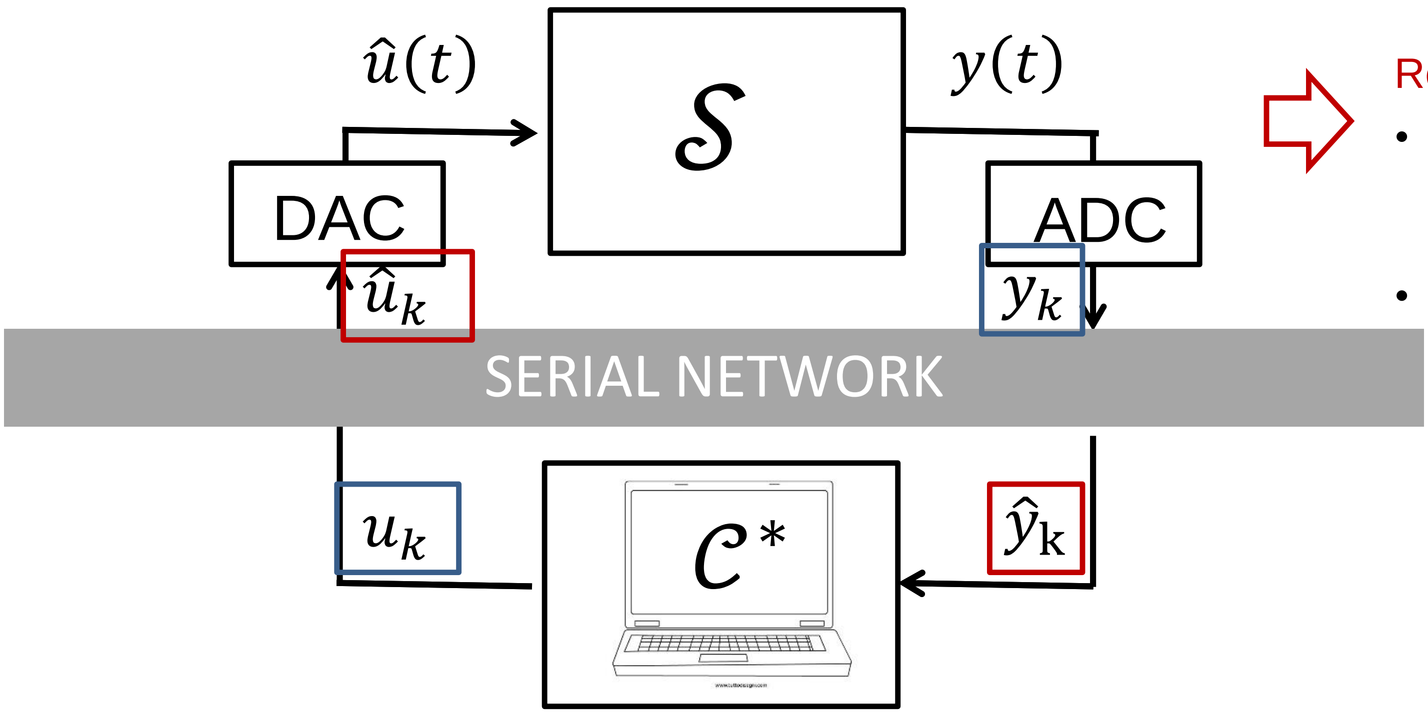

Requirements:

- Bounded – and as constant as possible – time delay
- Guaranteed transmission

Glossary

- **Medium**: physical channel on which communication takes place.
- **Node**: an entity connected to the network (e.g.., sensor, actuator, controller). In networked control applications each node must be «smart», i.e., it must include
  - a transceiver for communicating
  - a processor that runs the application program
- **Control network / control bus**: system for interconnecting nodes. It refers commonly to **hardware** (wires,...) **and software** (e.g., error correction, protocols, addressing) **attributes** of the communication medium.
- **Interoperability**: the ability of products of different manufacturers to work together.

Glossary

- **Response time**: time that takes for an event detected by one node to cause an action to occur in another node of the network.
- **Packet:** piece of information passed, consisting of status and command information.
- **Offered traffic**: amount of information that a network **is required** to carry at a given time.
- **Throughput**: amount of information actually **carried by the network** at a given time.
- **MAC**: **m**edia **a**ccess **c**ontrol function defined by the IEEE 802 commitee for local area networks.

Delays in control networks

Time delay:

$$T_{\text{delay}} = T_{\text{dest}} - T_{\text{src}}$$

- T_{src}: instant when the source node begins the process of sending a message.
- T_{dest}: instant when the destination node completes the reception.

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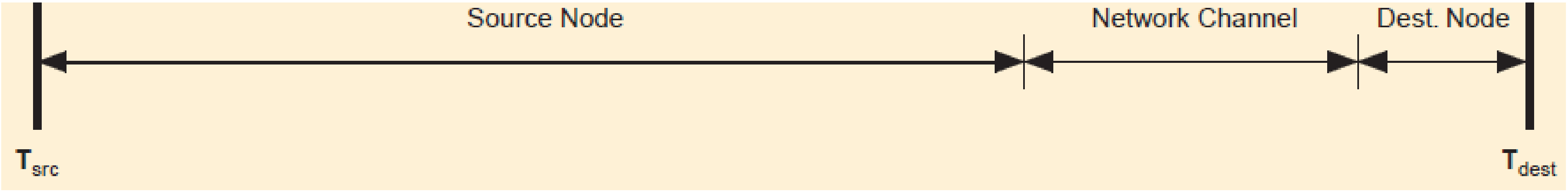
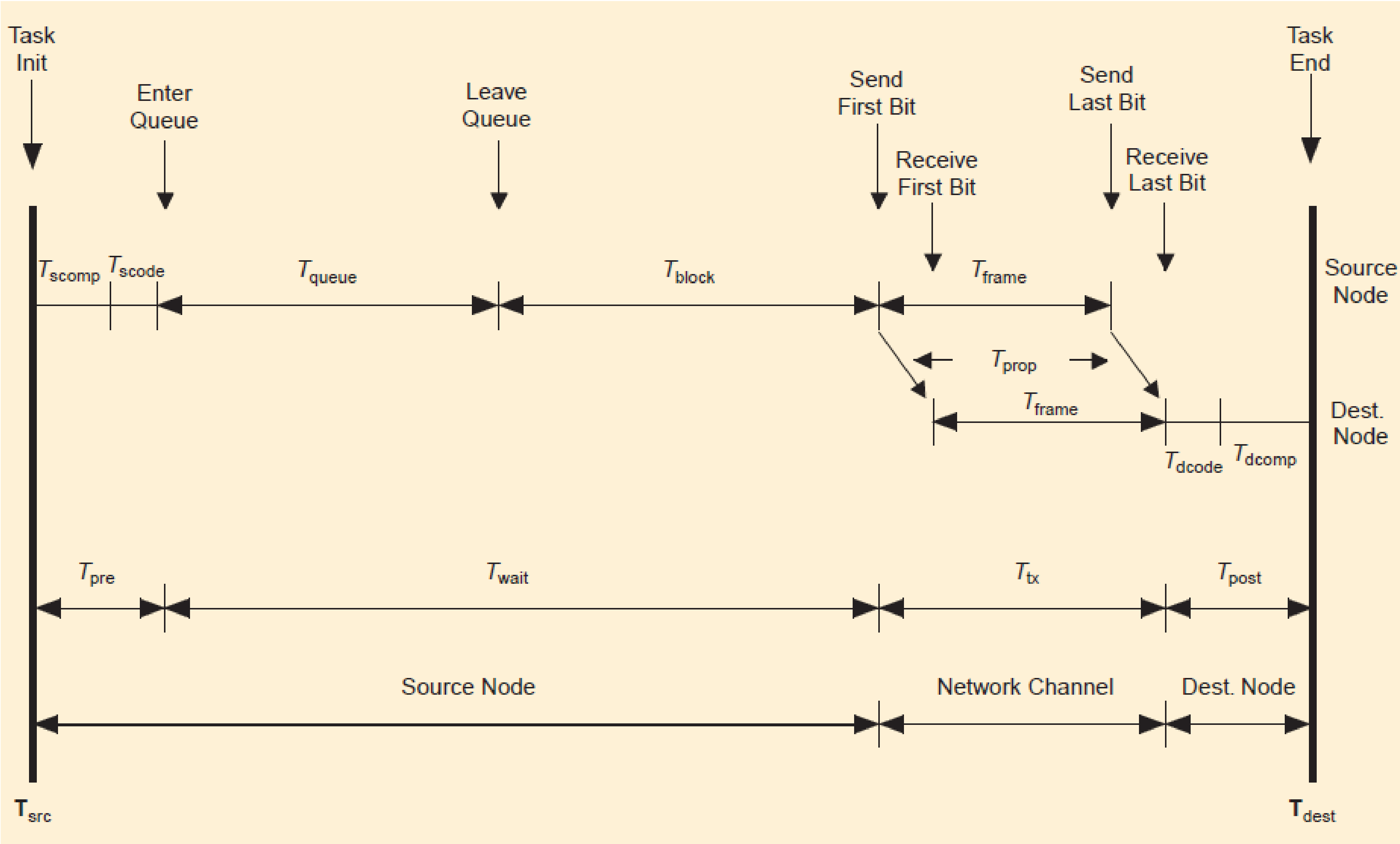
The time delay T_{delay} is composed of three parts:

- Time delay at the **source node**
- Time delay on the **network channel**
- Time delay at **destination**



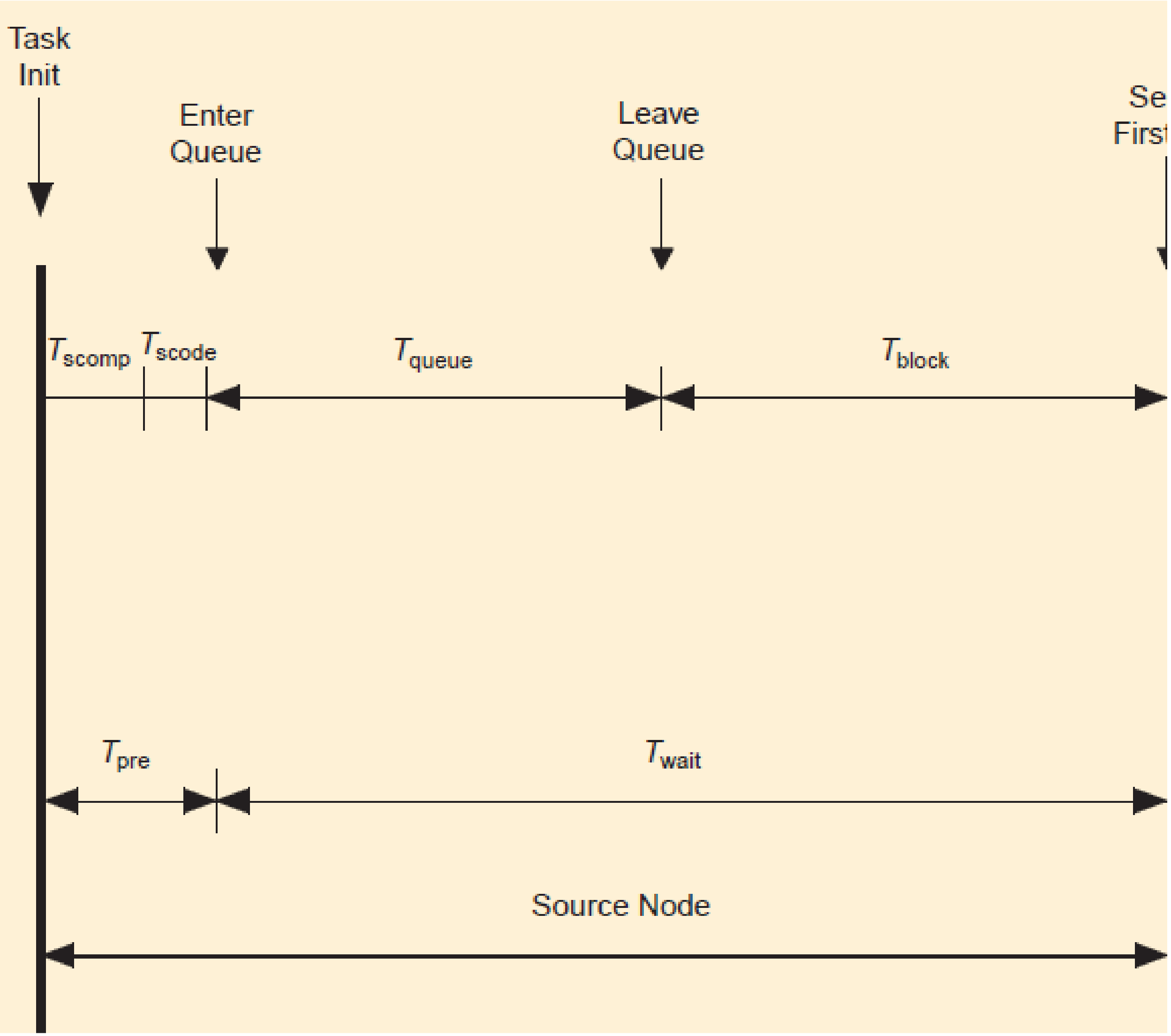
Delays in control networks



From: F-L Lian, J.R.Moyne, D M Tilbury, Performance Evaluation of control networks: Ethernet, ControlNet, DeviceNet. *IEEE Control System Magazine*, 21(1), 2001, pp. 66-83

Delays in control networks

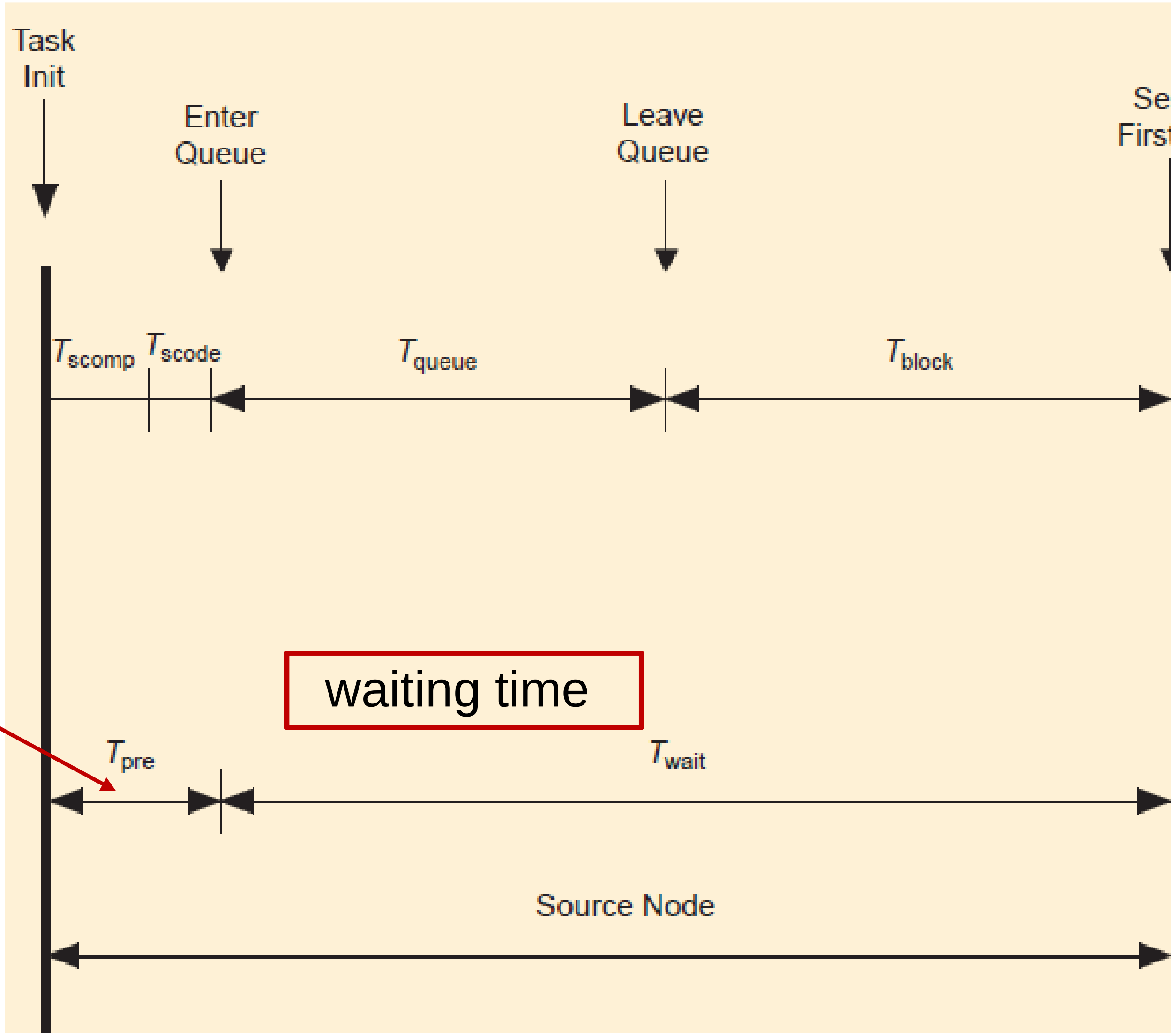
Time delay at source



Delays in control networks

Time delay at source

processing time



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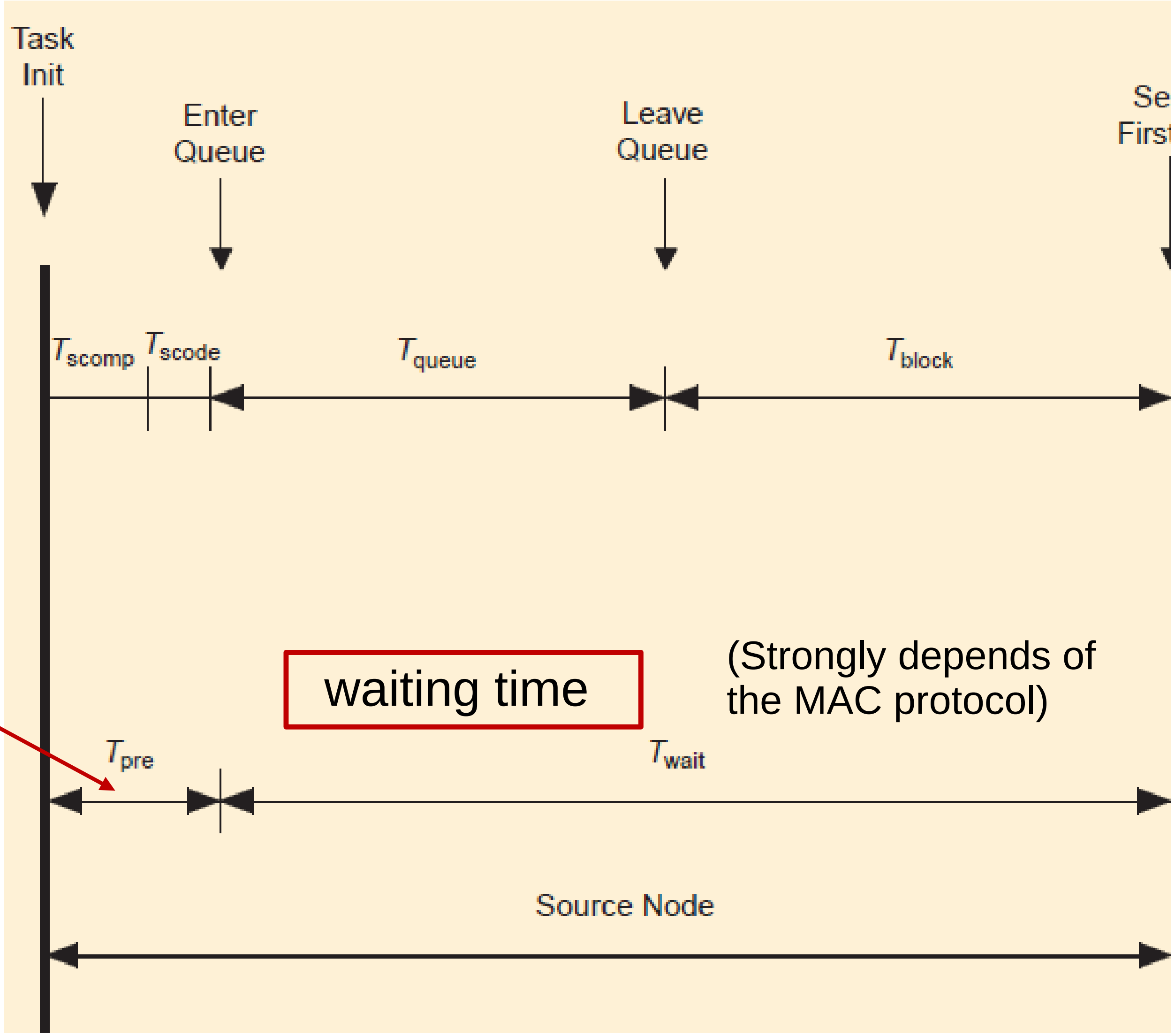

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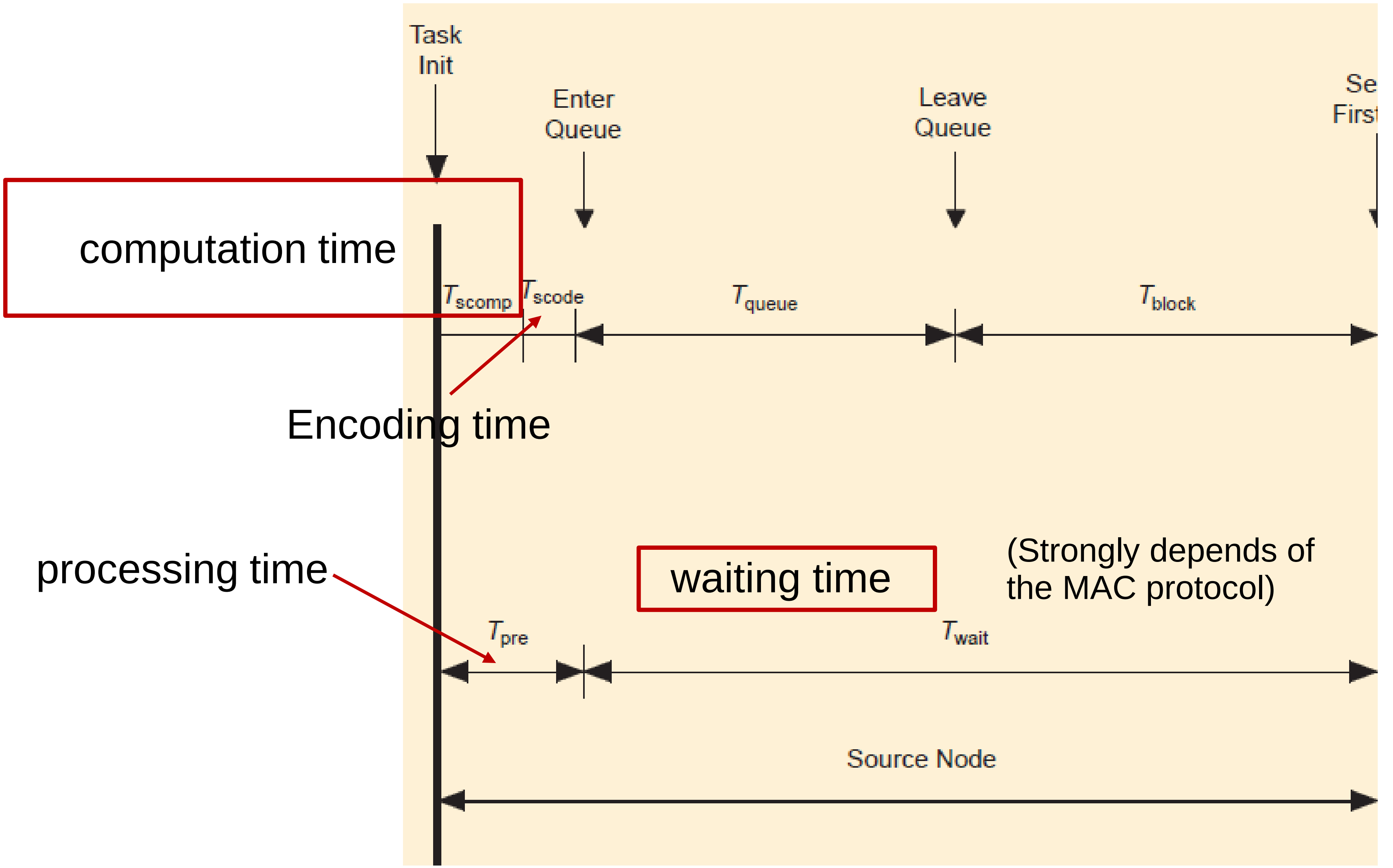

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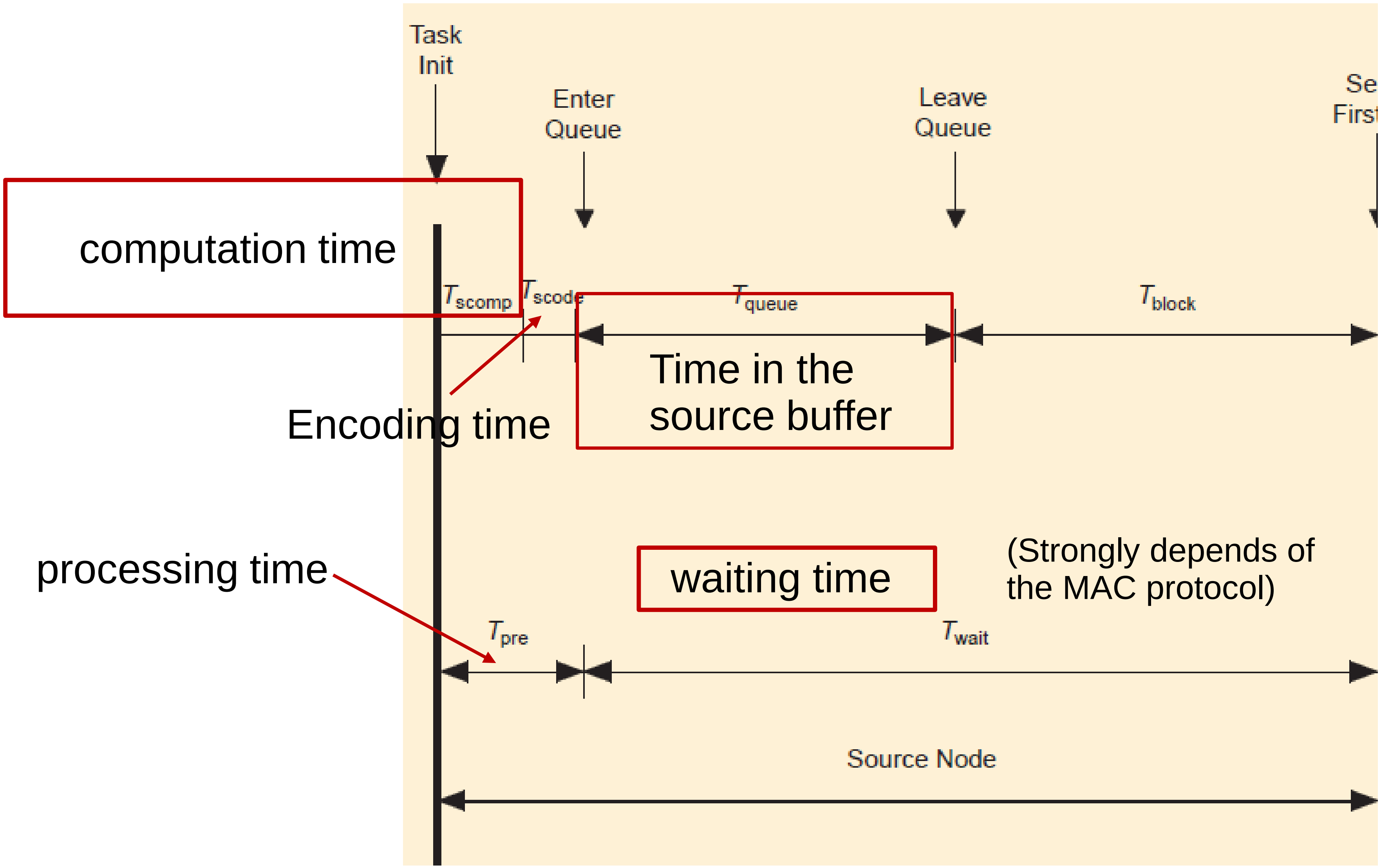

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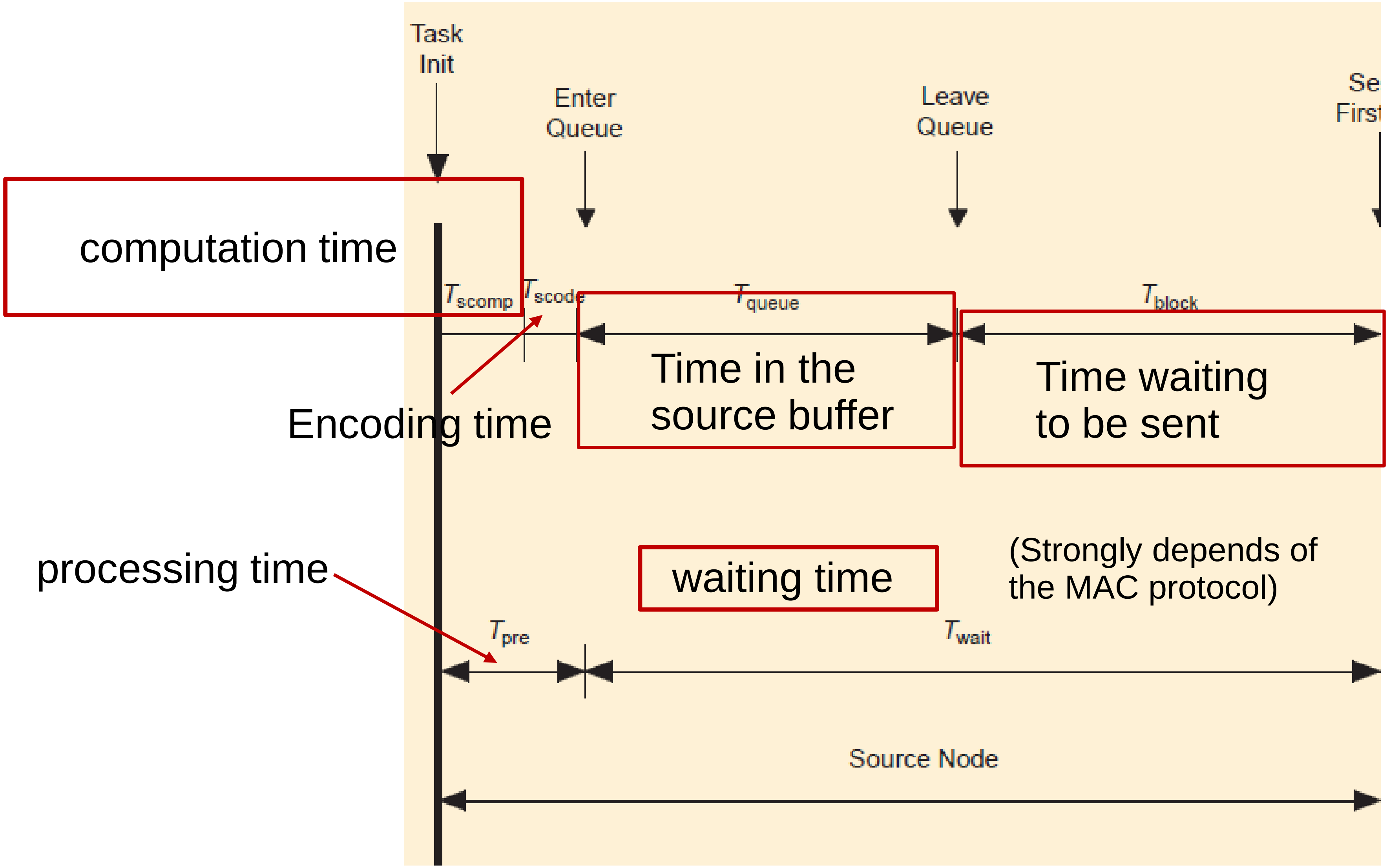

Delays in control networks

Time delay on the network channel

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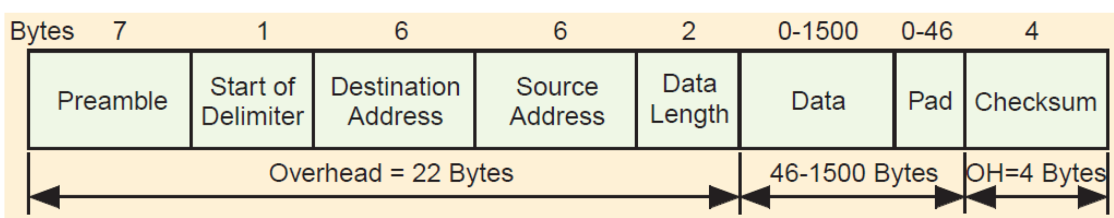
In fact, control networks are **packet** (frame) **networks**: a whole packet is sent «at the same time», consisting of header + data, e.g. (for Ethernet)



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i.e., the max. number of bits per second [b/s] transmissible by the link

Remark: S/N is commonly expressed in dB ($|x|_{dB} = 10 \log_{10} |x|$), but in Shannon's equation it is used in linear scale.

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Remark: The time delay on the network channel is often negligible with respect to other time delays (especially the waiting time).

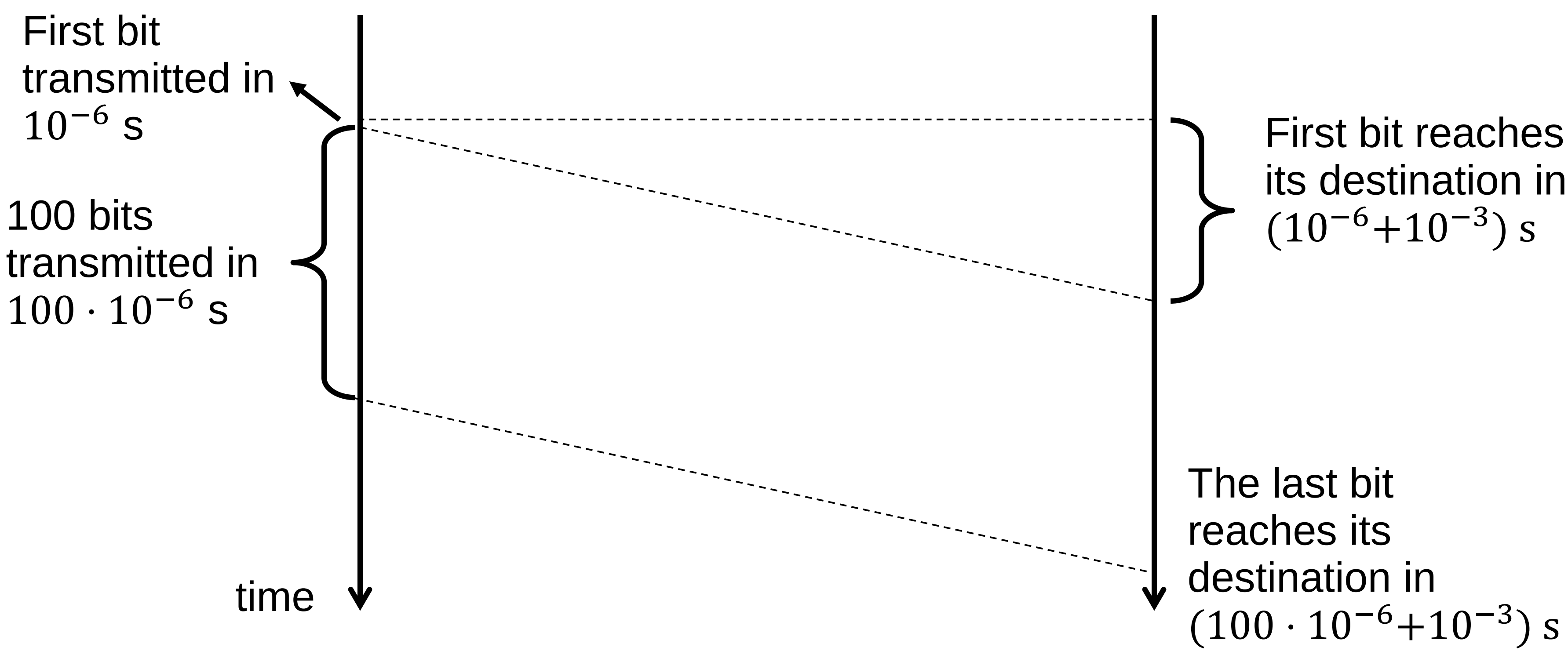
Delays in control networks

Time delay on the network channel

Example: we need to transmit a 100 bits packet on a link having

- R=1 Mb/s
- T_{prop}=1 ms.

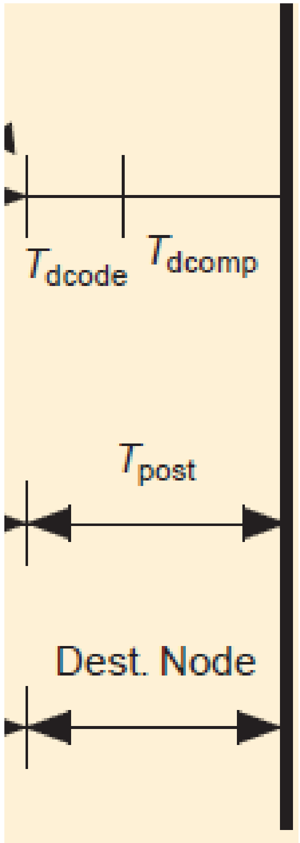
$$\Rightarrow \quad T_{\text{tx}} = 10^{-3} + \frac{100}{1000000} = 1.1 \cdot 10^{-3}\ s$$



Delays in control networks

Time delay at destination

Post-processing time



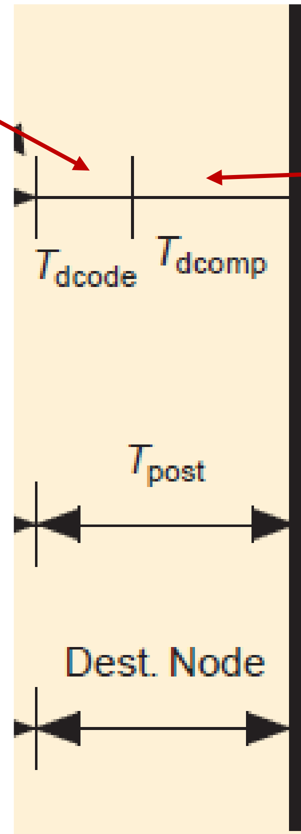
Delays in control networks

Time delay at destination

Decoding time

Computation time

Post-processing time



Delays in control networks

- **Network efficiency**:

$$P_{\text{eff}} = \frac{T_{\text{tx}}}{T_{\text{delay}}}$$

- **Network utilization**:

$$P_{\text{util}} = \frac{\text{Total time used to transmit data}}{\text{total running time}}$$

If P_{util} is low, there is bandwidth left for other purposes. If P_{util} is large (i.e., close to 1), the network is saturated.

- **Stability of a network**: if the number of unsent messages in the queue of each node is greater or equal to a constant or diverges, then the network is unstable.

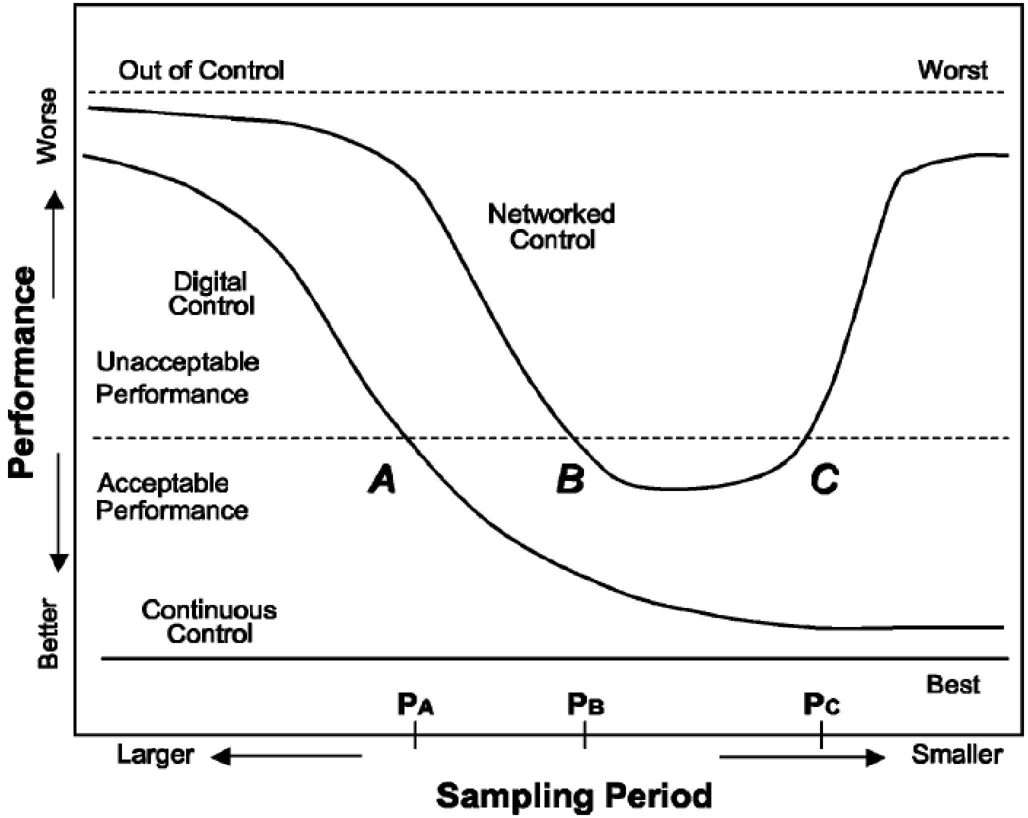
Remark that

- Stability of the network
- Stability of the networked control system

are different concepts.
Of course, if the network is unstable, the network-induced delays may be large enough to induce instability of the network control system!

Delays in control networks

The choice of the sampling time is particularly critical.



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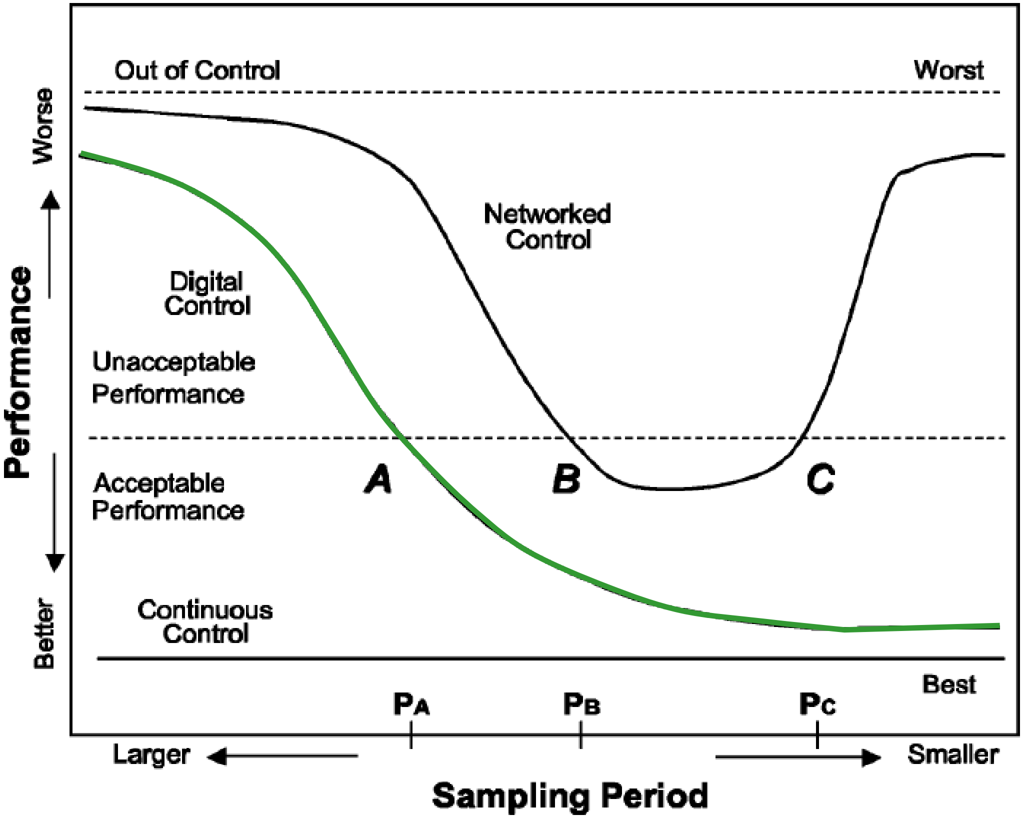

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Delays in control networks

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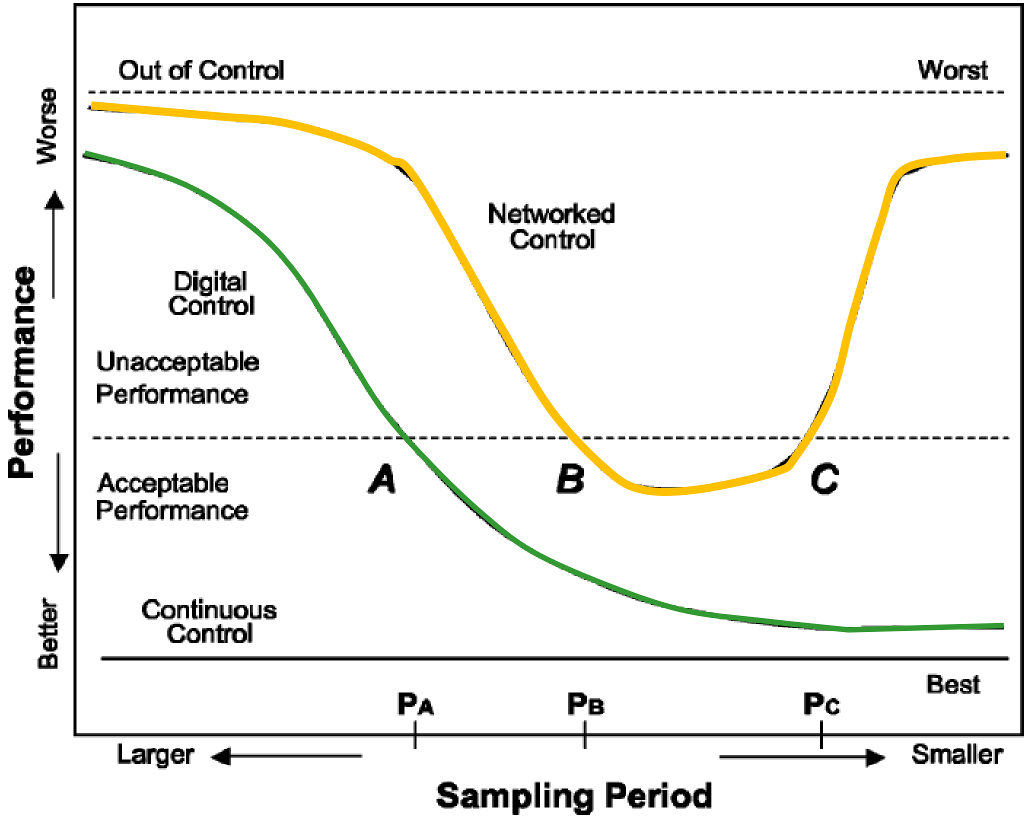
- In classical **digital control systems**, the smaller the sampling period, the smaller the induced phase lag -> the better performance
- In **networked control systems**, the smaller the sampling period, the more information is required to be sent across the control network per time unit. Below a certain value
  - The network gets congested
  - Networked-induced delays become serious
  - The phase lag induced by the network-induced delays become predominant



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Control network protocols

A **control network** is defined by the MAC protocol + physical layer specifications.

For example:

- ControlNet
- DeviceNet
- Ethernet (including Wireless Ethernet)
- PROFIBUS, etc...

CAN, instead, is just a protocol.

There are two main **classes of MAC protocols**:

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There are two main **classes of MAC protocols**:

1. The ones based on **scheduling**, i.e., time division multiplexing, where access time is allocated in a round-robin fashion (alternating in a «regular way») among the nodes:
   1. Token passing, TP (token bus/token ring)
   2. Master-slave, MS
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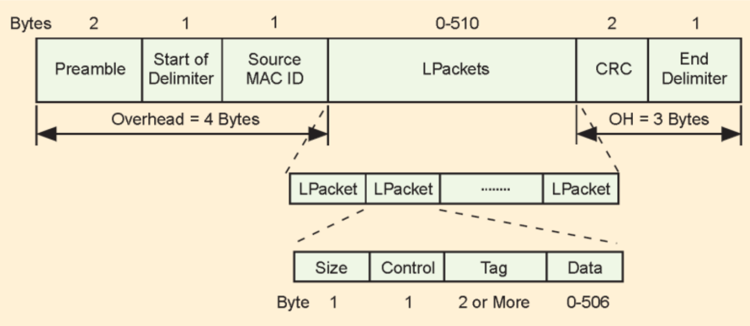
2. The ones based on **random multiple access** (CSMA) protocols:
    - With collision arbitration (CSMA/CA), CAN.
    - With no collision arbitration (Ethernet), e.g., collision detection (CSMA/CD).

Control network protocols

Time-division multiplexing

Token-passing (TP) methods (PROFIBUS and ControlNet).

- The node with the token is allowed to send data.
- When it finishes to send data or the token holding time expires, the node «passes» the token to its next (logical or physical) neighbor.
- In ControlNet, each node can hold the token for 22.4/827 μs
- In **token bus** methods the token is passed through a logical ring, while in **token rings** methods the token is passed through a physical ring.



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(destination is specified in Tag)

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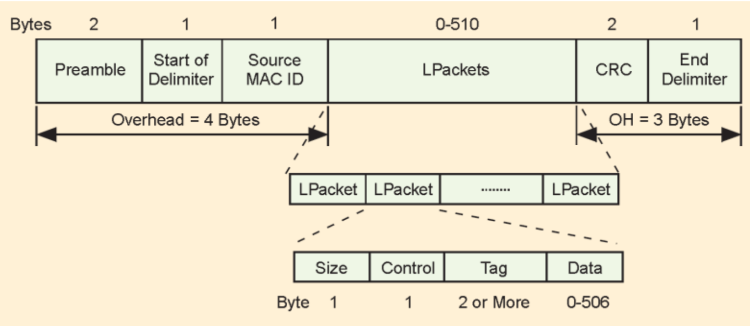

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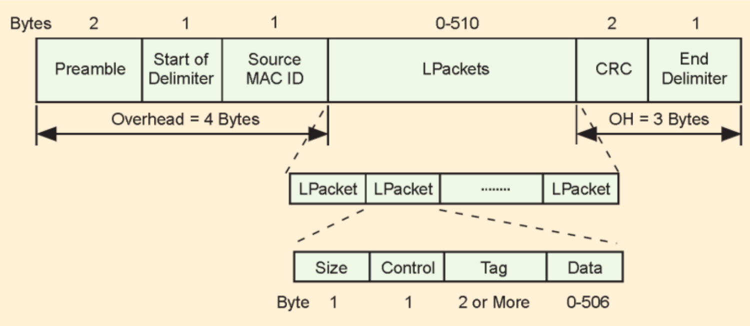

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Time division multiple access methods (used in firewire)

1. It has a transmission cycle of 125 μs divided into time slots.
2. Each node is guaranteed a transmission timeslot at each cycle.

Control network protocols

Time-division multiplexing

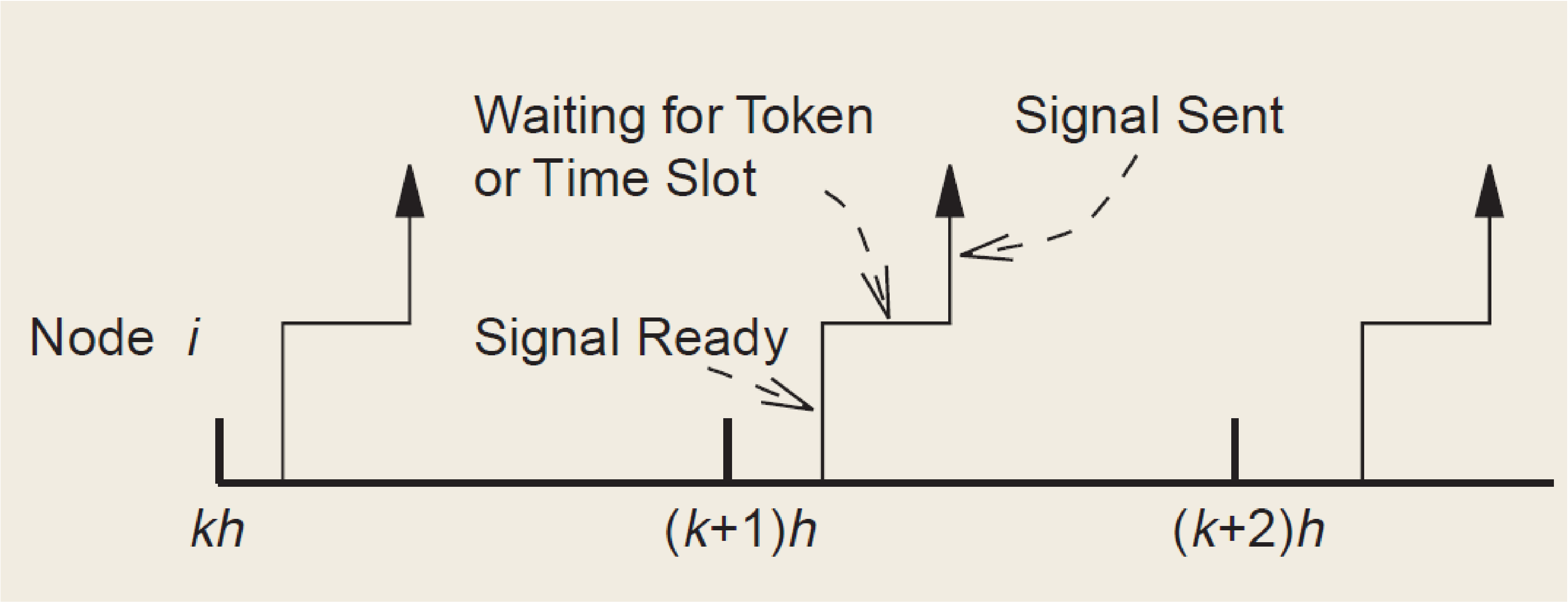
- **PROS**:
  - Deterministic protocols.
  - The maximum waiting time before sending a message can be characterized by the maximum «rotation time».
  - No time wasted in **collision** and **contention** for network resources.
  - No single node can monopolize the network.
  - Excellent throughput and efficiency at large network loads.
  - Nodes can be added or removed during the network operation for most of these protocols.

- **CONS**:
  - Poor performance at low traffic loads
  - When there are many nodes in a logical ring and when the traffic is light, a large percentage of the network time is used in passing the token.

Control network protocols

Time-division multiplexing

Timing diagram for a node



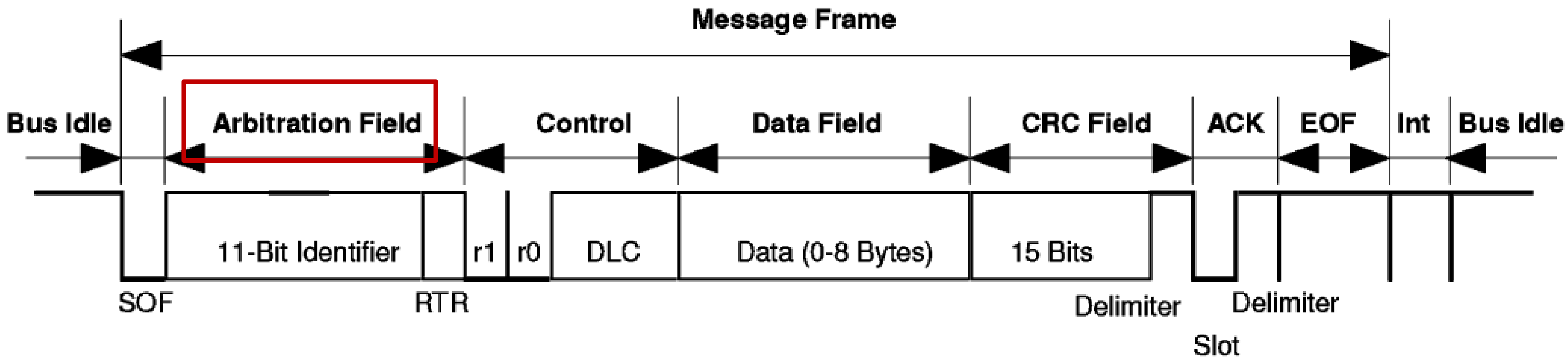
From: W. Zhang, M.S.Branicky, S.M.Philips. Stability of networked control systems, *IEEE Control Systems Magazine*, 21(1), 2001, pp. 84-99.

Control network protocols

Random access with collision arbitration (CAN, used by DeviceNet)

CAN is optimized for short messages and uses a CSMA/AMP (Carrier sense multiple access with Arbitration on Message Priority) medium access method.

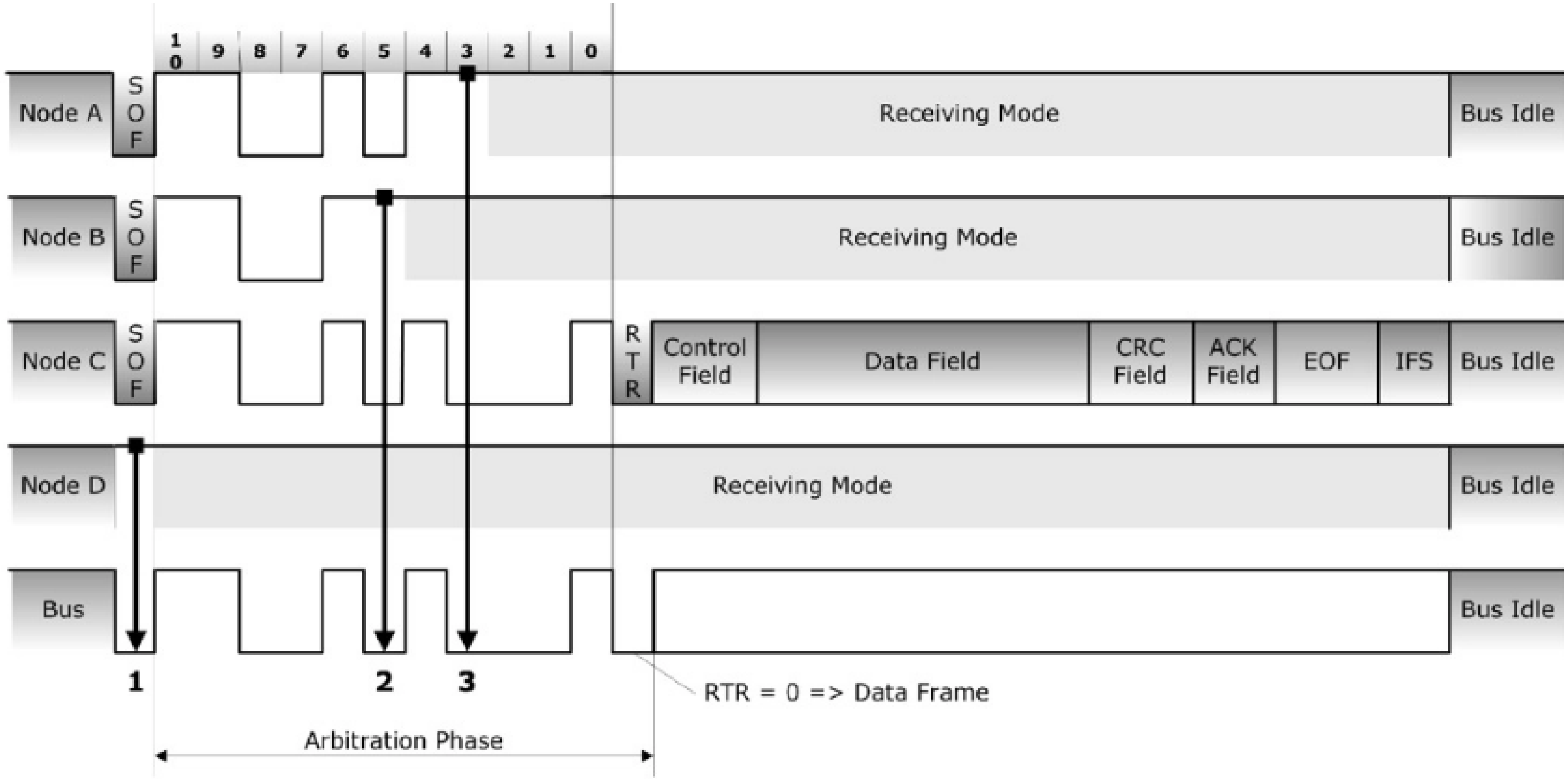
- Each message has a specific prority that is used to arbitrate access to the bus in case of simulataneous transmission,



- A node that wants to transmit a message waits until the bus is free and then starts to send the identifier of the message bit by bit.
- Conflicts for access to the bus resolved during transmission by an arbitration process thanks to the arbitration field.
- If two devices want to send a message at the same time, they send the message bit by bit and then listen to the network: if one receives a bit different from the one that has sent out, he loses the right to continue sending.

Control network protocols

Random access with collision arbitration (CAN, used by DeviceNet)



Node D is not trying to transmit.
Node C wins the bus access within 12 clock times.

At a baud rate of 1 MBit/sec this would translate into a 12 μs delay!

Control network protocols

Random access with collision arbitration (CAN, used by DeviceNet)

- Overhead: 47 bits, including 11-bit arbitration field.
- Data packet frame size: 0-8 bytes.
- Data rate: 500kb/s (for a 100m network).

PROS:

- Optimized for short messages.
- Priority is specified in the arbitration field.
- Larger priority messages always gain access to the medium during arbitration.
- Transmission delay (for high-priority messages) can be guaranteed.

CONS:

- Relatively slow data rate.
- Throughput is limited compared to other control networks.
- CAN is not suitable for the transmission of messages with large data sizes, although it supports fragmentation.

Control network protocols

Random access with no collision arbitration (Ethernet)

Ethernet uses a CSMA medium access method.

- A node that wants to send a message waits until the bus is free.
- When the bus is free, the node waits an interframe time + backoff time (to reduce the probability of collisions. The interframe time may depend on some priorities) and then starts to send a message.
- A collision occurs when two nodes start to send exactly at the same time.
- There are different types («flavours») of Ethernet, e.g.:
  - Hub-based (CSMA/CD, i.e., with collision detection).
  - Wireless Ethernet.

Control network protocols

Hub-based Ethernet (CSMA/CD, i.e., with collision detection)

- If nodes detect a collision when sending, they stop transmitting and then wait a random length of time before to start sending again.
- After 10 collision, the retry time is fixed.
- After 16 collision, the node stops attempting.

- Overhead: 26 bytes.
- Data packet frame size: 46-1500 bytes.
- Data rate: 10Mb/s (common in control networks).
- Max length: 2500 m.

PROS:
- Almost no delays at low network loads.
- No bandwidth is used to gain access to the network (compared to token-based networks).

CONS:
- It is non-deterministic and does not support message prioritization.
- At high network load, collisions cause possibly unbounded delays.
- A message may be discarded after a number of collision: end-to-end communication is not guaranteed.
- Ethernet may have a relatively large message size to convey small amounts of data.

Control network protocols

Wireless Ethernet (simplified)

- Besides the physical layer, the largest difference with the «wired» version, lies in the MAC protocol.
- Uses interframe spaces and varying backoff times to prevent collisions, similarly to the CSMA/CD protocol.
- Wireless stations cannot «hear» the collisions.

⇨ A collision avoidance mechanism is used but cannot completely prevent collisions

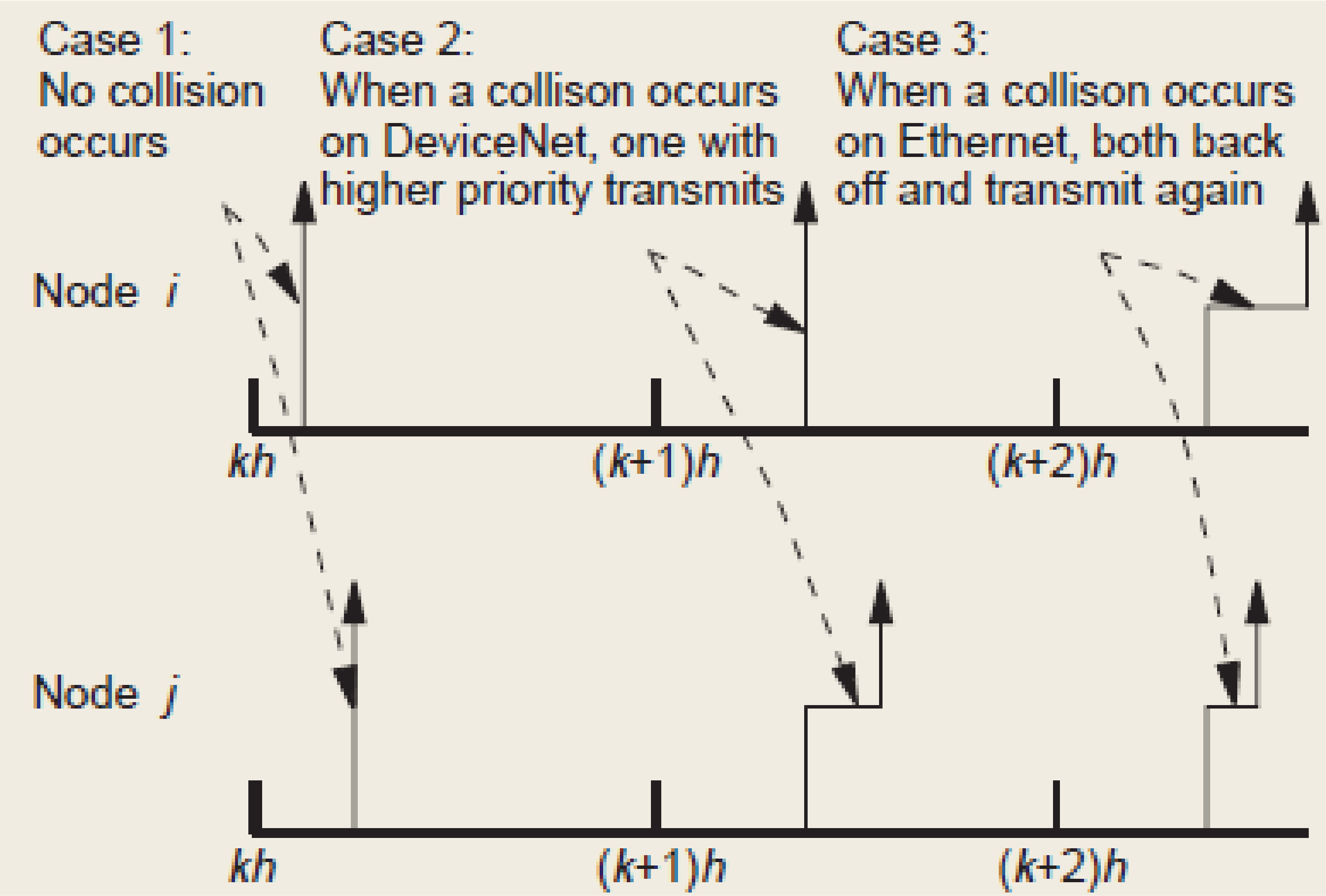
- After the destination node successfully receives a packet, it send the **ACKnowledgement packet** back to the sender. If the sender does not receive the ACK, it assumes that the transmission was unsuccessful and retransmits.

- Raw data rates range from 11 to 54 Mb/s
- The actual throughput is lower, due to
  - Overhead associated with interframe delays, ACKs.
  - Actual implementations of the network adapter.

Control network protocols

Random access methods

Timing diagram for two nodes

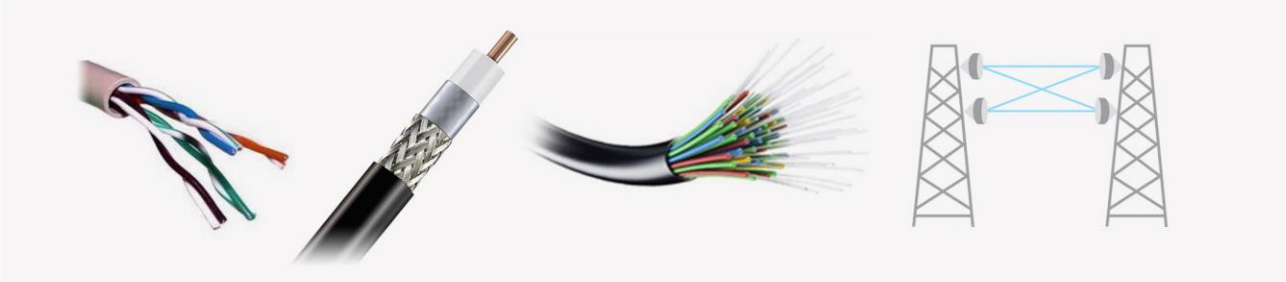


From: W. Zhang, M.S.Branicky, S.M.Philips. Stability of networked control systems, *IEEE Control Systems Magazine*, 21(1), 2001, pp. 84-99.

Control network types

	ETHERNET LAN	DEVICENET (CAN-based)	CONTROLNET	WIRELESS ETHERNET IEEE 802.11
MAC	CSMA/CD	CSMA/AMP	TDM/TP (Token held for 22.4- 827 μs)	CSMA
Bitrate	10Mb/s –10Gb/s Standard: 1 Gb/s	125, 250, 500 kb/s	5 Mb/s	11 Mb/s (IEEE 802.11b) - 54 Mb/s (IEEE 802.11g)
MAX Nodes (devices)	1024 (Expandable with routers)	64	99	??
Distance	100 m – 50 km (fiber with switches), typical 2.5 km	100-500 m	1000 m	35 m (indoor), 140 m (outdoor), up to 5000 m (IEEE 802.11a)
MIN message size	46 bytes	0 bytes	0 bytes	0 bytes
MAX message size	1500 bytes	8 bytes	506 bytes	2312 bytes
overhead	26 bytes	47 bits, including 11 bits arbitration field	7 bytes + 4 bytes for each Lpacket	34 bytes
Medium	Twisted pair, thin coaxial, and optical fiber	Thin/thick round cable, flat cable	Coaxial cable, optical fiber	Vacuum (infra-red and radio frequency spectrum waves) NB: endpoints are typically connected with wired infrastructures
% Users	78%	47%	39%	35%

Control network media



Media	% of c	Description
Thick coaxial cable	77%	Originally used for ethernet, referred to as "thicknet"
Thin coaxial cable	65%	Referred to as ethernet "thinnet" or "cheapernet"
Unshielded twisted pair	59%	Multipaired copper cabling used for LAN and telecom applications
Microstrip	57%	PCB trace on FR4 dielectric, μr = 3.046
Stripline	47%	PCB trace in FR4 dielectric, μr = 4.6
Optical fiber	67%	Silica waveguide used to transport optical energy
Vacuum	100%	Vacuum or free space

From: J. Coffey, Latency in optical fiber systems. *COMMSCOPE*.

Packet dropouts in control networks

Packet dropouts occur in case of

- **Failures of nodes or physical network links** (the latter is more common in wireless than wired networks)

- **Buffer overflows** during congestions.

- **Message collisions** and/or unacceptable delays:

    - Although many protocols include a message-retry mechansim, this can be done for a limited time. After this time expires, the message is dropped.

    - In network control systems it may be advisable to discard old unsent data in favor or newly available information (e.g., control signals of measurements).

Main related references

- R.S. Raji, Smart networks for control. *IEEE Spectrum*, 31 (6), 1994, pp. 49-55.
- W. Zhang, M.S.Branicky, S.M.Philips. Stability of networked control systems, *IEEE Control Systems Magazine*, 21(1), 2001, pp. 84-99.
- F-L Lian, J.R.Moyne, D M Tilbury, Performance Evaluation of control networks: Ethernet, ControlNet, DeviceNet. *IEEE Control System Magazine*, 21(1), 2001, pp. 66-83.
- J. R. Moyne and D. M. Tilbury. The emergence of industrial control networks for manufacturing control, diagnostics, and safety data. *Proceedings of the IEEE*, 95 (1), pp. 29-47